

Introduction to CPS:

- Key features of CPS: reactive computation, concurrency, feedback control of the physical world, real time computation, safety critical applications
- General structure of a CPS.
- Examples of CPS.

IoT:

- Characteristics of IoT
- Enabling Technologies
- Concept of Transducers/sensors-Primary & Secondary, active & passive, analog & digital, Concept about actuators-thermal, electric & mechanical
- IoT stack
- Levels of IoT

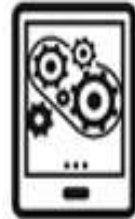
IoT Stack

	IOT STACK	WEB STACK
<i>TCP/IP</i>	<i>IOT applications Device Management</i>	<i>Web applications</i>
<i>Data Format</i>	<i>Binary, JSON, CBOR</i>	<i>HTML, XML, JSON</i>
<i>Application Layer</i>	<i>CoAP, MQTT, XMPP, AMPQP</i>	<i>HTTP, DHCP, DNS, TLS/SSL</i>
<i>Transport Layer</i>	<i>UDP, DTLS</i>	<i>TCP, UDP</i>
<i>Internet Layer</i>	<i>IPv6/IP Routing</i> <i>6LOWPAN</i>	<i>IPv6, IPv4, IPSec</i>
<i>Network/Link Layer</i>	<i>IEEE 802.15.4 MAC</i> <i>IEEE 802.15.4 PHY / Physical Radio</i>	<i>Ethernet (IEEE 802.3), DSL, ISDN, Wireless LAN (IEEE 802.11), Wi-Fi</i>

The IoT Technology Stack



Device
Hardware



Device
Software



Communications



Cloud
Platform

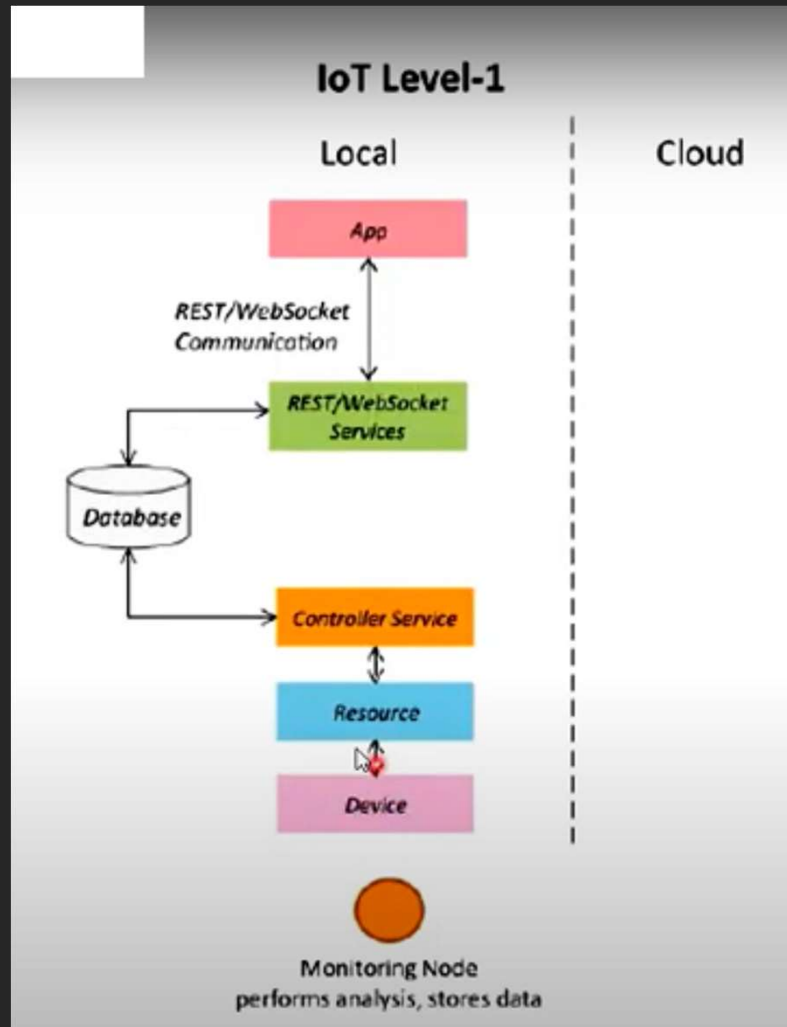


Cloud
Applications

IoT Layers

IoT system comprises of following workings:

- Device
- Resource
- Controller Service
- Database
- Web Service
- Analysis Component
- Application



Level_1

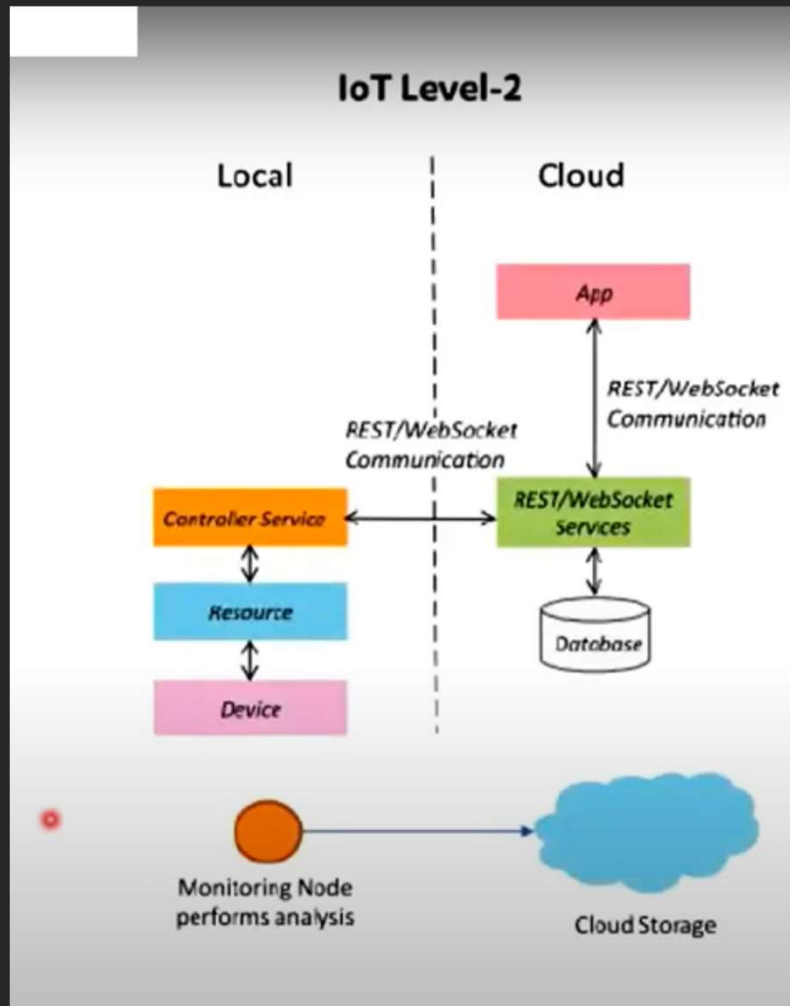
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- A level-1 IoT system has a single node/device that performs sensing and/or actuation, stores data, performs analysis and hosts the application
- Level-1 IoT systems are suitable for modeling low cost and low-complexity solutions where the data involved is not big and the analysis requirements are not computationally intensive

Level_1

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Level_2

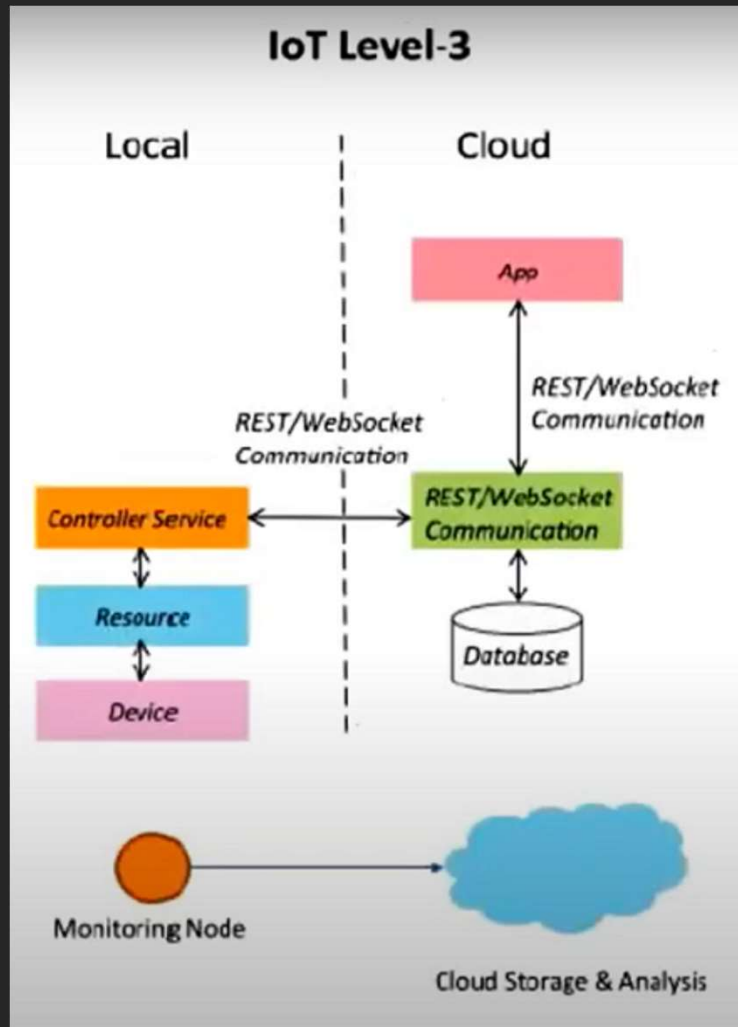


- Level-2 IoT systems are suitable for solutions where the data involved is big

- A level-2 IoT system has a single node that performs sensing and/or actuation and local analysis
- Data is stored in the cloud and application is usually cloud based

Level_2

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Level_3

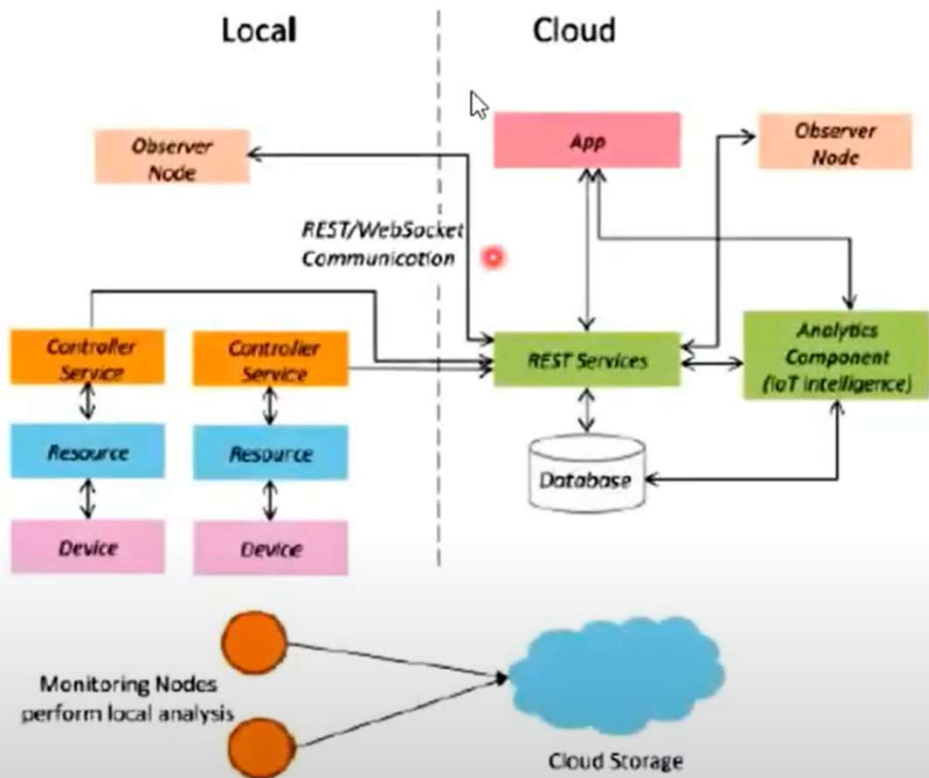
- A level-3 IoT system has a single node. Data is stored and analyzed in the cloud and application is cloud based

- Level-3 IoT systems are suitable for solutions where the data involved is big and the analysis requirements are computationally intensive

Level_3

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Level_4

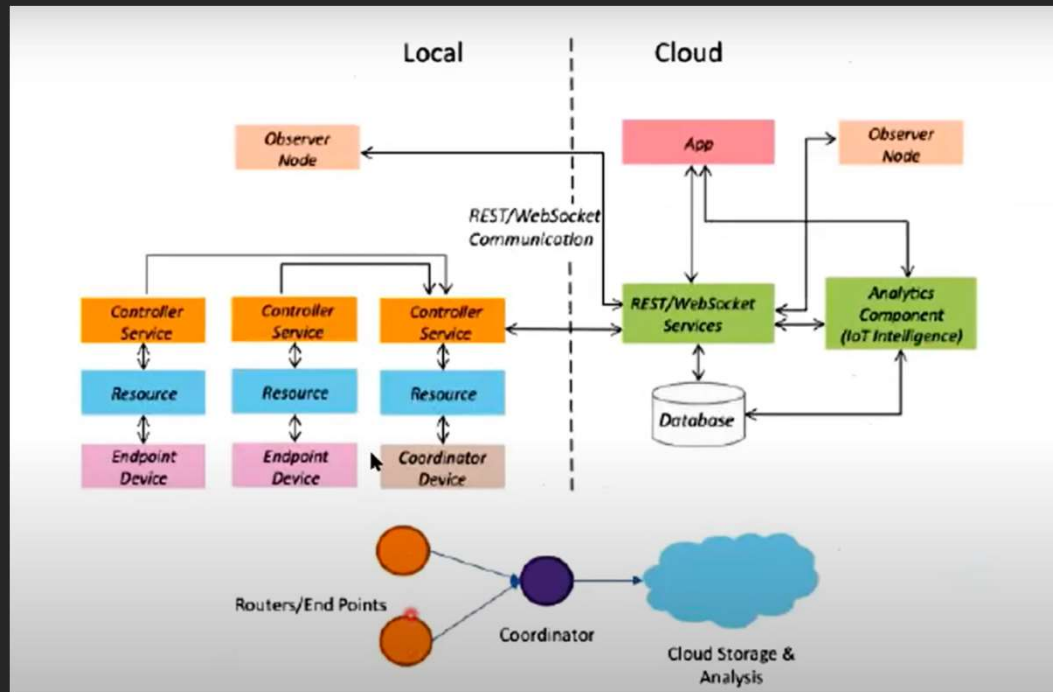


- A level-4 IoT system has multiple nodes that perform local analysis. Data is stored in the cloud and application is cloud-based

- Level-4 contains local and cloud based observer nodes which can subscribe to and receive information collected in the cloud from IoT devices
- Level-4 IoT systems are suitable for solutions where multiple nodes are required 

Level_4

Level_5

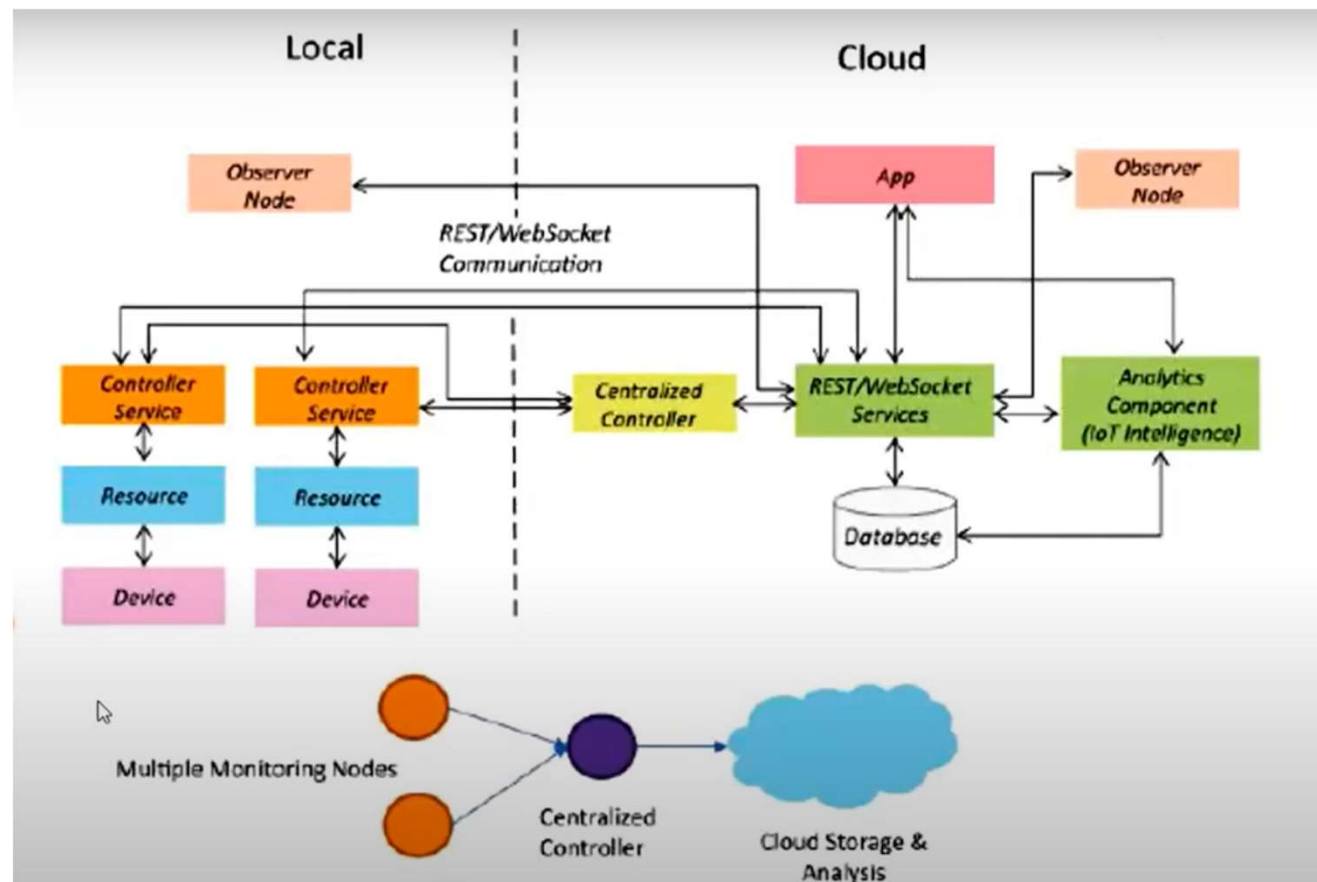


- A level-5 IoT system has multiple end nodes and one coordinator node
- The end nodes that perform sensing and/or actuation
- Coordinator node collects data from the end nodes and sends to the cloud

- Data is stored and analyzed in the cloud and application is cloud-based
- Level-5 IoT systems are suitable for solutions based on wireless sensor networks, in which the data involved is big and the analysis requirements are computationally intensive 

Level_5

Level_6



Level_6

- The results are visualized with the cloud-based application
- The centralized controller is aware of the status of all the end nodes and sends control commands to the nodes

- A level-6 IoT system has multiple independent end nodes that perform sensing and/or actuation and send data to the cloud
- Data is stored in the cloud and application is cloud-based
- The analytics component analyzes the data and stores the results in the cloud database